

Exercise: Production Rules – Chess Board Moves

Piece	First Move (YES/NO)	Front Empty (YES/NO)	Left Enemy (LE)	LE Protected	Right Enemy (RE)	RE Protected	Maximum Number of Steps	Direction	Movement Code	Time Stamp
	YES	YES	NO	-	NO	-	2	Forward	(0,2,0)	2024 Jan 1
	NO	YES	NO	-	NO	-	1	Forward	(0,1,0)	2024 Jan 1
	NO	-	YES	NO	-	-	1	Forward-L	(1,0,0)	2024 Jan 1
	NO	-	-	-	YES	NO	1	Forward-R	(0,0,1)	2024 Jan 1
	NO	YES	YES	YES	YES	NO	1	Forward-R	(0,0,1)	2024 Jan 1
	NO	YES	YES	YES	YES	YES	1	Forward	(0,1,0)	2024 Jan 1
	NO	YES	YES	NO	YES	YES	1	Forward-L	(1,0,0)	2024 Jan 1
	YES	YES	NO	-	NO	-	2	Forward	(0,-2,0)	2024 Jan 1
	NO	YES	NO	-	NO	-	1	Forward	(0,-1,0)	2024 Jan 1
	NO	-	YES	NO	-	-	1	Forward-L	(-1,0,0)	2024 Jan 1
	NO	-	-	-	YES	NO	1	Forward-R	(0,0,-1)	2024 Jan 1
	NO	YES	YES	YES	YES	NO	1	Forward-R	(0,0,-1)	2024 Jan 1

	NO	YES	YES	YES	YES	YES	1	Forward	(0,-1,0)	2024 Jan 1
	NO	YES	YES	NO	YES	YES	1	Forward-L	(-1,0,0)	2024 Jan 1
	NO	YES	YES	YES	YES	YES	1	Forward-R	(0,0,1)	2024 Feb 15
	NO	YES	YES	YES	YES	YES	1	Forward-R	(0,0,-1)	2024 Feb 15

Given the Knowledge Table Above, Do the following requirements:

1. Represent All the given knowledge using Production Rules.
2. Use the following Working Memories and execute them on the Production Rules, and check the output, it must be matching the Table:



c.



d.



e.



3. In the given knowledge there are conflicts. Highlight them, and what will be the output of the Inference Engine when those cases appear? Explain your answer.